Extra Challenges

Finished early? Pick any of these. None of them need components you do not already have.

#	Challenge	Uses
1	Make an LED fade smoothly instead of blinking	analogWrite, ~ pins
2	Build a two-road junction: when one is green, the other is red	6 LEDs
3	A lamp that blinks faster the darker it gets	LDR + delay
4	A reversing alarm: the buzzer beeps faster as your hand comes closer	HC-SR04 + buzzer
5	A servo that follows your hand: closer hand, bigger angle	HC-SR04 + servo
6	Count how many times a hand passes the sensor in one minute	HC-SR04 + Serial
7	A doorbell that plays a two-tone chime	switch + buzzer
8	A night lamp that also beeps when it becomes very dark	LDR + buzzer
9	A traffic signal with a night caution mode that blinks only yellow	traffic module
10	A toll gate that shows the money collected, not just the count	Serial + maths

MIXED REVISION

Twenty Questions to Test Yourself

1. What are the three steps every robot follows?
2. Why must a circuit be a complete loop?
3. Which leg of an LED goes towards the plus side?
4. What happens if you connect an LED with no resistor?
5. How many holes are joined together in one breadboard column?
6. What is the difference between a digital pin and an analog pin?
7. What does void setup() do that void loop() does not?
8. Write the line of code that switches pin 7 on.
9. How many milliseconds are there in three seconds?
10. What are the first two settings to check before uploading?
11. Why does the traffic module need only four wires?
12. What is the largest number analogRead() can give?
13. What does calibration mean?
14. Why do we print sensor values before writing an if - else?
15. Explain what a threshold is, in one sentence.
16. How does the HC-SR04 know how far away something is?
17. Why does the distance formula divide by two?
18. What is the difference between a servo motor and an ordinary DC motor?
19. Why do programmers write their own instructions like readDistance()?
20. Name one thing you would add to your toll booth with one more session.

Words to Know

Word	Meaning
Actuator	Any part that moves or acts: a motor, a buzzer, a lamp.
Analog	A value that can be anything in a range, not just on or off.
Breadboard	A block of holes for joining parts without soldering.
Calibration	Adjusting a machine to suit the real place it works in.
Compile	Turning typed words into instructions the chip can follow.
Current	How much electricity flows every second.
Debugging	Hunting down the reason something does not work.
Digital	A value that is only ON or OFF, HIGH or LOW.
Echolocation	Finding distance by timing an echo. Bats do it; so does the HC-SR04.
Flowchart	A picture of what a machine will do, step by step.
Function	An instruction you write yourself and then reuse.
Ground (GND)	The return path. Every circuit needs one.
IDE	The software window where you write and upload your program.
LDR	Light Dependent Resistor - a sensor that feels brightness.
LED	Light Emitting Diode - a lamp that works only one way round.
Loop	Code that repeats again and again.
Millisecond	One thousandth of a second.
Module	A small ready-made board with components already fitted.
Resistor	A part that limits current, measured in ohms.
Sensor	A part that turns something physical into a number.
Servo	A motor that turns to an exact angle and holds it.
Threshold	The number where a decision changes from one answer to the other.
Upload	Sending your program from the laptop into the board.
Voltage	How hard electricity is being pushed, measured in volts.

Answer Key

Section A only. Sections B and C are for you and your trainer to discuss - there is often more than one good answer.

Check Yourself - Unit 1

Q	Answer	The correct option
1	(b)	act
2	(c)	a complete loop
3	(b)	protect it
4	(a)	columns of five
5	(c)	the plus side
6	(c)	resistance

Check Yourself - Unit 2

Q	Answer	The correct option
1	(b)	once at the start
2	(b)	switch pin 9 on
3	(c)	half a second
4	(b)	board and port
5	(c)	analog pins
6	(b)	a semicolon

Check Yourself - Unit 3

Q	Answer	The correct option
1	(b)	resistors are built in
2	(b)	not work at all
3	(c)	5 seconds
4	(b)	before coding
5	(b)	the delay value after green
6	(b)	slow down safely

Check Yourself - Unit 4

Q	Answer	The correct option
1	(c)	0 and 1023
2	(b)	A0
3	(b)	the board can talk to the laptop
4	(b)	from real readings
5	(b)	make a choice
6	(b)	a number

Check Yourself - Unit 5

Q	Answer	The correct option
1	(b)	timing an echo
2	(b)	sound travels there and back
3	(b)	turn to 90 degrees
4	(b)	an instruction we wrote
5	(b)	counting the same car twice
6	(b)	and

My Project Log

Fill one row after every class. This is your evidence, and it counts towards your assessment.

Session	What I built or learnt	A problem I hit	How I fixed it
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			

A last word

Nothing in this book worked the first time when we built it either. The wires came loose, the servo trembled, and the counter once counted a single car four times. Every one of those faults taught us something. Keep the notebook, keep asking why, and next year you will be building things this book cannot teach you.